Geometry & Measurement

ANSWER KEY

Angle facts

1 Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

Angles on a straight line sum to 180°: $3x + 2x + 30 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$.

2 Find the angle vertically opposite a 65° angle.

Vertically opposite angles are equal, so the angle is 65°.

3 Three angles around a point measure 110°, 95°, and x. Find x.

Angles around a point sum to 360°: $x = 360 - 110 - 95 = 155^\circ$.

4 Two angles are complementary (sum to 90°). One measures 37°. Find the other.

$90 - 37 = 53^\circ$.

Parallel lines and transversals

5 Two parallel lines are cut by a transversal, creating an angle of 70°. Find its corresponding angle.

Corresponding angles are equal, so the angle is 70°.

6 In the same setup, find the co-interior (allied) angle to a 70° angle.

Co-interior angles sum to 180°: $180 - 70 = 110^\circ$.

7 Find the alternate interior angle to a 55° angle.

Alternate interior angles are equal, so the angle is 55°.

8 Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

Co-interior angles sum to 180°: $4x + 2x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$.

Triangle angle sum

9 Find the third angle of the triangle shown, given the two marked angles.

$180 - 40 - 75 = 65^\circ$.

10 An isosceles triangle has two equal angles of 72° each. Find the third angle.

$180 - 72 - 72 = 36^\circ$.

11 In a right triangle, one of the non-right angles is 35°. Find the other non-right angle.

$180 - 90 - 35 = 55^\circ$.

12 A triangle has three equal angles. Find each angle, and name this type of triangle.

$180 \div 3 = 60^\circ$ each — this is an equilateral (equiangular) triangle.

Angle sums of polygons

13 Find the sum of the interior angles of a hexagon (6 sides).

Sum $= (n - 2) \times 180^\circ = (6 - 2) \times 180^\circ = 720^\circ$.

14 Find the sum of the interior angles of a pentagon (5 sides).

Sum $= (5 - 2) \times 180^\circ = 540^\circ$.

15 A regular polygon has an interior angle sum of $1{,}080^\circ$. How many sides does it have?

$(n - 2) \times 180 = 1{,}080 \Rightarrow n - 2 = 6 \Rightarrow n = 8$ sides (a regular octagon).

The Pythagorean theorem

16 Find the hypotenuse of the right triangle shown.

$c = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ cm.

17 A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

$\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

18 A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

$\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ m.

19 A right triangle has legs 9 and 12. Find the hypotenuse.

$\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

Perimeter and area of rectangles and squares

20 A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

Perimeter $= 2(12 + 7) = 38$ cm. Area $= 12 \times 7 = 84$ cm^2.

21 A square has side length 9 cm. Find its perimeter and area.

Perimeter $= 4 \times 9 = 36$ cm. Area $= 9^2 = 81$ cm^2.

22 A rectangle has perimeter 50 cm and length 16 cm. Find its width and area.

Width $= \frac{50}{2} - 16 = 25 - 16 = 9$ cm. Area $= 16 \times 9 = 144$ cm^2.

23 A square has area $121\ \text{cm}^2$. Find its side length and perimeter.

Side $= \sqrt{121} = 11$ cm. Perimeter $= 4 \times 11 = 44$ cm.

Area of triangles and parallelograms

24 A triangle has base 14 cm and height 9 cm. Find its area.

Area $= \frac{1}{2} \times 14 \times 9 = 63\ \text{cm}^2$.

25 A parallelogram has base 12 cm and height 8 cm. Find its area.

Area $= 12 \times 8 = 96\ \text{cm}^2$.

26 A triangle has area $60\ \text{cm}^2$ and base 15 cm. Find its height.

Height $= \frac{2 \times 60}{15} = 8$ cm.

27 A parallelogram has area $126\ \text{cm}^2$ and height 9 cm. Find its base.

Base $= \frac{126}{9} = 14$ cm.

Circles: circumference and area

28 A circle has radius 7 cm. Find its circumference, using $\pi = \frac{22}{7}$.

$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm.

29 A circle has radius 7 cm. Find its area, using $\pi = \frac{22}{7}$.

$A = \pi r^2 = \frac{22}{7} \times 49 = 154\ \text{cm}^2$.

30 A circle has diameter 20 cm. Find its circumference in terms of π.

$C = \pi d = 20\pi$ cm.

31 Find the area of the shaded 90° sector shown, in terms of π.

Sector area $= \frac{90}{360} \times \pi \times 10^2 = \frac{1}{4} \times 100\pi = 25\pi\ \text{cm}^2$.

Composite figures

32 Find the area of the L-shaped room shown.

Treat it as a $10\ \text{m} \times 8\ \text{m}$ rectangle with a $4\ \text{m} \times 3\ \text{m}$ notch removed: $10 \times 8 - 4 \times 3 = 80 - 12 = 68\ \text{m}^2$.

33 A garden is L-shaped: a 15 m by 6 m rectangle with a 5 m by 2 m rectangle cut from one corner. Find its area.

$15 \times 6 - 5 \times 2 = 90 - 10 = 80\ \text{m}^2$.

34 A shape consists of a 12 cm by 5 cm rectangle with a right triangle of base 6 cm and height 5 cm attached to one of the short sides. Find the total area.

Rectangle: $12 \times 5 = 60$ cm^2. Triangle: $\frac{1}{2} \times 6 \times 5 = 15$ cm^2. Total $= 60 + 15 = 75$ cm^2.

Volume and surface area

35 A rectangular box has dimensions 5 cm $\times 4$ cm $\times 3$ cm. Find its volume.

$V = 5 \times 4 \times 3 = 60$ cm^3.

36 A cube has edge length 6 cm. Find its volume and surface area.

$V = 6^3 = 216$ cm^3. Surface area $= 6 \times 6^2 = 6 \times 36 = 216$ cm^2.

37 A rectangular prism has volume 240 cm^3, length 10 cm, and width 6 cm. Find its height.

Height $= \frac{240}{10 \times 6} = \frac{240}{60} = 4$ cm.

38 Find the surface area of a rectangular box with dimensions 8 cm $\times 5$ cm $\times 4$ cm.

$SA = 2(lw + lh + wh) = 2(40 + 32 + 20) = 2 \times 92 = 184$ cm^2.

Word problems

39 A rectangular garden measuring 20 m by 15 m is surrounded by a path 1 m wide. Find the area of the path alone.

Outer dimensions: 22 m $\times 17$ m $= 374$ m^2. Garden area: $20 \times 15 = 300$ m^2. Path area $= 374 - 300 = 74$ m^2.

40 New flooring costs 45 riyals per square meter. Find the total cost to floor a rectangular room measuring 6 m by 4 m.

Area $= 6 \times 4 = 24$ m^2. Cost $= 24 \times 45 = 1{,}080$ riyals.

41 A 17 m ladder leans against a vertical wall with its base 8 m from the wall. How high up the wall does it reach?

$\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$ m.

42 A jogger runs 4 laps of a circular track of radius 35 m, using $\pi = \frac{22}{7}$. Find the total distance covered.

One lap $= 2\pi r = 2 \times \frac{22}{7} \times 35 = 220$ m. Four laps $= 4 \times 220 = 880$ m.

MCQ — Angle facts

43 Two angles are supplementary. One is three times the other. Find the smaller angle.

a. 60°

b. 30°

c. 135°

d. 45° ✓

$x+3x=180 \Rightarrow x=45^{\circ}$.

44 Three angles around a point measure 118°, 142°, and x. Find x.

a. 80°

b. 140°

c. 120°

d. 100° ✓

$x=360-118-142=100^{\circ}$.

45 An angle is four times its complement. Find the angle.

a. 54°

b. 18°

c. 72° ✓

d. 90°

$x=4(90-x) \Rightarrow 5x=360 \Rightarrow x=72^{\circ}$.

46 Two angles on a straight line measure $(4x+10)^{\circ}$ and $(2x+20)^{\circ}$. Find x.

a. 20

b. 35

c. 15

d. 25 ✓

$4x+10+2x+20=180 \Rightarrow 6x+30=180 \Rightarrow x=25$.

MCQ — Parallel lines and transversals

47 Two parallel lines are cut by a transversal. One angle is $(3x+15)^\circ$ and its corresponding angle is $(5x-25)^\circ$. Find x.

a. 10

b. 8

c. 30

d. 20 ✓

Corresponding angles are equal: $3x+15=5x-25 \Rightarrow x=20$.

48 Two parallel lines cut by a transversal create co-interior angles of $(2x+10)^\circ$ and $(3x-30)^\circ$. Find x.

a. 30

b. 40 ✓

c. 50

d. 34

Co-interior angles sum to 180°: $2x+10+3x-30=180 \Rightarrow 5x=200 \Rightarrow x=40$.

49 Two parallel lines are cut by a transversal. One angle is 63°. Find its alternate exterior angle.

a. 153°

b. 27°

c. 63° ✓

d. 117°

Alternate exterior angles are equal, so the answer is 63°.

50 Two parallel lines are cut by a transversal. One angle and its co-interior angle differ by 40° and sum to 180°. Find the larger angle.

a. 110° ✓

b. 70°

c. 140°

d. 90°

$L + S = 180, L - S = 40 \Rightarrow L = 110^\circ$.

MCQ — Triangle angle sum

51 Find the third angle of the triangle shown.

a. 116°

b. 64°

c. 54°

d. 44° ✓

$180 - 64 - 72 = 44^\circ$.

52 The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.

a. 55°

b. 70°

c. 40° ✓

d. 110°

$180 - 70 - 70 = 40^\circ$.

53 A triangle has three angles in the ratio $2:3:4$. Find the largest angle.

a. 90°

b. 100°

c. 80° ✓

d. 60°

9 parts $=180^\circ$, so 1 part $=20^\circ$; largest $=4\times 20=80^\circ$.

54 In a right triangle, one non-right angle exceeds the other by 26°. Find the larger non-right angle.

a. 32°

b. 58° ✓

c. 64°

d. 48°

$x+(x+26)=90 \Rightarrow x=32$; larger $=58^\circ$.

MCQ — The Pythagorean theorem

55 Find the hypotenuse of the right triangle shown.

a. 23

b. 19

c. 17 ✓

d. 13

$\sqrt{8^2+15^2}=\sqrt{64+225}=\sqrt{289}=17$.

56 Find the missing leg of the right triangle shown.

a. 26

b. 18

c. 24 ✓

d. 20

$\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24.$

57 A rectangular field is 24 m long and 18 m wide. Find the length of its diagonal.

a. 30 ✓

b. 36

c. 42

d. 26

$\sqrt{24^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30$ m.

58 A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?

a. Only if it is also isosceles

b. **Yes — a right triangle** ✓

c. Cannot be determined

d. No — not a right triangle

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$, so yes, it is a right triangle.

59 A 15 m ladder leans against a wall with its foot 9 m from the wall's base. How high up the wall does the ladder reach?

a. 13

b. 10

c. 12 ✓

d. 18

$\sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144} = 12$ m.

MCQ — Perimeter and area of rectangles and squares

60 Find the perimeter and area of the rectangle shown.

a. $P = 48$ **cm,** $A = 135$ **cm**^{2} ✓

b. $P = 48$ cm, $A = 24$ cm^2

c. $P = 135$ cm, $A = 48$ cm^2

d. $P = 24$ cm, $A = 135$ cm^2

$P = 2(15 + 9) = 48$ cm; $A = 15 \times 9 = 135$ cm^2.

61 Find the perimeter and area of the square shown.

a. $P = 44$ cm, $A = 110$ cm^2

b. $P = 44$ **cm,** $A = 121$ **cm**^{2} ✓

c. $P = 22$ cm, $A = 121$ cm^2

d. $P = 121$ cm, $A = 44$ cm^2

$P = 4 \times 11 = 44$ cm; $A = 11^2 = 121$ cm^2.

62 A rectangle has perimeter 64 cm and length 20 cm. Find its width and area.

a. $w = 12$ cm, $A = 32$ cm^2

b. $w = 12$ **cm,** $A = 240$ **cm**^{2} ✓

c. $w = 44$ cm, $A = 240$ cm^2

d. $w = 22$ cm, $A = 440$ cm^2

$2(l + w) = 64 \Rightarrow l + w = 32 \Rightarrow w = 12$ cm; $A = 20 \times 12 = 240$ cm^2.

63 A square has area 225 cm^2. Find its side length and perimeter.

a. $s = 15$ cm, $P = 30$ cm

b. $s = 225$ cm, $P = 900$ cm

c. $s = 15$ **cm,** $P = 60$ **cm** ✓

d. $s = 45$ cm, $P = 60$ cm

$s = \sqrt{225} = 15$ cm; $P = 4 \times 15 = 60$ cm.

MCQ — Area of triangles and parallelograms

64 The triangle shown has base 16 cm and height 10 cm. Find its area.

a. 160

b. 26

c. 80 ✓

d. 40

$A = \frac{1}{2}(16)(10) = 80 \text{ cm}^2$.

65 The parallelogram shown has base 18 cm and height 7 cm. Find its area.

a. 252

b. 126 ✓

c. 63

d. 25

$A = 18 \times 7 = 126 \text{ cm}^2$.

66 A triangle has area $84\ \text{cm}^2$ and height 12 cm. Find its base.

a. 14 ✓

b. 1,008

c. 28

d. 7

$b = \dfrac{2(84)}{12} = 14$ cm.

67 A parallelogram has area $150\ \text{cm}^2$ and base 15 cm. Find its height.

a. 20

b. 10 ✓

c. 2,250

d. 5

$h = \dfrac{150}{15} = 10$ cm.

MCQ — Circles: circumference and area

68 Find the circumference of the circle shown, using $\pi = \dfrac{22}{7}$.

a. 44

b. 616

c. 88 ✓

d. 176

$C = 2 \times \dfrac{22}{7} \times 14 = 88$ cm.

69 Find the area of the circle shown, using $\pi = \frac{22}{7}$.

a. 132

b. 2,772

c. 693

d. 1,386 ✓

$A = \frac{22}{7} \times 21^2 = 1{,}386$ cm^2.

70 Find the circumference of the circle shown, in terms of π.

a. 48π cm

b. 24π **cm** ✓

c. 12π cm

d. 144π cm

$C = 2\pi r = 2\pi(12) = 24\pi$ cm.

71 Find the area of the shaded 60° sector shown, in terms of π.

a. 81π cm^2

b. 27π cm^2

c. 13.5π **cm**^{2} ✓

d. 4.5π cm^2

Sector area $= \frac{60}{360} \times \pi(9)^2 = 13.5\pi$ cm^2.

72 A circle has area 144π cm^2. Find its radius.

a. 6

b. 24

c. 12 ✓

d. 72

$r^2 = 144 \Rightarrow r = 12$ cm.

MCQ — Composite figures

73 Find the area of the L-shaped room shown (all measurements in meters).

a. 70

b. 58 ✓

c. 46

d. 82

Full rectangle $10 \times 7 = 70$ minus the 4×3 notch $= 70 - 12 = 58$ m^2.

74 A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.

a. 108

b. 126 ✓

c. 162

d. 138

$18 \times 8 = 144$; $144 - 6 \times 3 = 144 - 18 = 126$ m^2.

75 Find the area of the shape shown: a 14 cm by 6 cm rectangle with a right triangle of base 6 cm and height 4 cm attached to one side.

a. 108

b. 90

c. 96 ✓

d. 72

Rectangle $= 14 \times 6 = 84$; triangle $= \frac{1}{2}(6)(4) = 12$; total $= 96$ cm^2.

76 A rectangular lawn 20 m by 12 m has a circular flower bed of radius 3.5 m removed from it. Find the remaining lawn area, using $\pi = \frac{22}{7}$.

a. 196

b. 278.5

c. 240

d. 201.5 ✓

Rectangle $= 240$; circle $= \frac{22}{7}(3.5)^2 = 38.5$; remaining $= 240 - 38.5 = 201.5\ \text{m}^2$.

MCQ — Volume and surface area

77 Find the volume of the rectangular box shown.

a. 120 ✓

b. 60

c. 15

d. 30

$V = 6 \times 5 \times 4 = 120\ \text{cm}^3$.

78 Find the volume and surface area of the cube shown.

a. $V = 343\ \text{cm}^3$, $SA = 49\ \text{cm}^2$

b. $V = 49\ \text{cm}^3$, $SA = 294\ \text{cm}^2$

c. $V = 294\ \text{cm}^3$, $SA = 343\ \text{cm}^2$

d. $V = 343$ **cm**^{3}**,** $SA = 294$ **cm**^{2} ✓

$V = 7^3 = 343\ \text{cm}^3$; $SA = 6 \times 7^2 = 294\ \text{cm}^2$.

79 Find the volume of the cylinder shown, using $\pi = \frac{22}{7}$.

a. 1,540 ✓

b. 770

c. 440

d. 3,080

$V = \frac{22}{7}(7)^2(10) = 1{,}540\ \text{cm}^3$.

80 Find the volume of the cone shown, using $\pi = \frac{22}{7}$.

a. 770 ✓

b. 2,310

c. 1,540

d. 385

$V = \frac{1}{3} \times \frac{22}{7}(7)^2(15) = 770\ \text{cm}^3$.

81 A rectangular prism has volume $360\ \text{cm}^3$, length 12 cm, and width 5 cm. Find its height.

a. 6 ✓

b. 15

c. 72

d. 30

$h = \frac{360}{12 \times 5} = 6$ cm.

MCQ — Coordinate geometry: distance and midpoint

82 Find the distance between the two points shown.

a. 5 ✓

b. 25

c. 7

d. $\sqrt{7}$

$\sqrt{(4-1)^2+(6-2)^2}=\sqrt{9+16}=\sqrt{25}=5.$

83 Find the midpoint of the segment joining the two points shown.

a. $(6,3)$

b. $(3,5)$

c. $(10,6)$

d. $(5,3)$ ✓

Midpoint $=\left(\frac{2+8}{2},\frac{3+3}{2}\right)=(5,3).$

84 Find the distance between the points $(0,0)$ and $(6,8)$.

a. 14

b. 10 ✓

c. 7

d. 48

$\sqrt{6^2+8^2}=\sqrt{36+64}=\sqrt{100}=10.$

85 Find the midpoint of the segment joining the two points shown.

a. $(-1,-4)$

b. $(4,1)$

c. $(2,8)$

d. $(1,4)$ ✓

Midpoint $=\left(\frac{-3+5}{2},\frac{1+7}{2}\right)=(1,4).$

MCQ — Similar figures and scale factor

86 Two similar triangles have a scale factor of $3:5$. If the smaller triangle's perimeter is 24 cm, find the larger triangle's perimeter.

a. 14.4

b. 72

c. 64

d. 40 ✓

$24 \times \frac{5}{3} = 40$ cm.

87 A model car is built at a scale of $1:24$. If the model is 8 cm long, find the actual car's length in meters.

a. 19.2

b. 0.192

c. 1.92 ✓

d. 24

$8 \times 24 = 192$ cm $= 1.92$ m.

88 Two similar rectangles have areas in ratio $4:9$. Find the ratio of their side lengths.

a. $4:9$

b. $2:3$ ✓

c. $8:18$

d. $16:81$

Side ratio $= \sqrt{4} : \sqrt{9} = 2:3$.

89 A photo is enlarged by a scale factor of 2.5. If the original photo is 6 cm wide, find the width of the enlargement.

a. 8.5

b. 12.5

c. 3.6

d. 15 ✓

$6 \times 2.5 = 15$ cm.

MCQ — Word problems

90 A rectangular pool measuring 25 m by 12 m is surrounded by a deck 2 m wide. Find the area of the deck alone.

a. 74

b. 300

c. 164 ✓

d. 464

Outer $= 29 \times 16 = 464$; pool $= 300$; deck $= 464 - 300 = 164$ m^2.

91 Tiling costs 38 riyals per square meter. Find the total cost to tile a rectangular room measuring 9 m by 6 m.

a. $1{,}026$ SAR

b. $4{,}104$ SAR

c. 342 SAR

d. $2{,}052$ **SAR** ✓

Area $= 9 \times 6 = 54$ m^2; cost $= 54 \times 38 = 2{,}052$ SAR.

92 A water tank shaped like the rectangular box shown has these dimensions in meters. Find its capacity in cubic meters.

a. 150

b. 60

c. 300 ✓

d. 21

$V = 10 \times 6 \times 5 = 300\ \mathrm{m}^3$.